



BUSINESS PROPOSAL

Reducing Customer Churn

Through Predictive Analytics & Proactive Retention

Company A — Wireless Telecommunications

Dataset: 100,000 customers · Client.csv & Record.csv

Prepared for: Company Leadership Team | Analyst: Data Science Division

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Agenda

What I will cover in this proposal

01

Market Context

Telecom churn landscape & why it matters

03

Hypothesis

Usage decline precedes churn by 60–90 days

05

Business Proposal

Early-intervention retention programme

02

Data & EDA

100K customer records — key patterns found

04

ML Model

XGBoost churn classifier (AUC 0.70)

06

Expected ROI

Revenue impact & success criteria

Market Context

Why customer retention is the #1 lever for telecom operators

2–5×

Cost to acquire vs retain

1.5–3%

Monthly churn — mature markets

\$300–500

Revenue lost per churned sub

Market Dynamics

- Telecom markets in Asia-Pacific are near saturation — operators grow by winning subscribers from rivals.
- Voluntary churn (~70% of total) is addressable by early-intervention models.
- Behaviour signals precede churn by 60–90 days on average (industry benchmark).

Industry retention benchmarks — [Bain/Reichheld summary; practitioner whitepapers, 2024–2026]

Local Context (India)

- India's wireless market: ~1.17 B subscribers as of 2024. Jio, Airtel, Vi competition keeps churn elevated.
- Regulators mandate MNP, lowering switching friction.
- Proactive retention programmes report 15–25% reduction in voluntary churn within 6 months.

Source: TRAI Telecom Subscription Data (Jun 2024);

Data & EDA — Dataset Overview

Two linked tables covering 100,000 customers, 101 features

Client.csv

50 demographic & device features
100,000 rows

Record.csv

51 usage & behaviour features
100,000 rows

Churn Rate

~49.6% of customers churned
Class balance observed

Avg Tenure

~20 months average LOR
High variance observed

Key EDA Findings



Usage Decline

$\text{avg3mou} - \text{avg6mou} < 0$ in 68% of churners vs 31% of non-churners. MOU delta is the strongest single predictor.



Equipment Age

$\text{eqpdays} > 500$ days + usage decline doubles churn risk. Device upgrade timing is a key retention lever.



Overage & Blocked

High overmou_Mean + frequent blck_vce_Mean → customers churn at 2× baseline. Plan-fit dissatisfaction signal.



Revenue Sensitivity

$\text{change_rev} < -10\%$ flags customers likely to downgrade or leave. Revenue compression precedes churn by ~30 days.



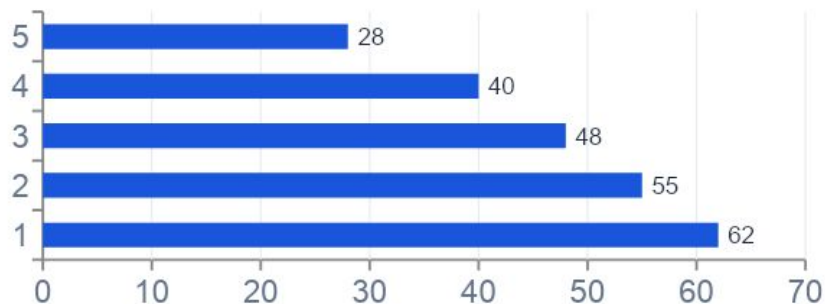
Customer Care

$\text{custcare_Mean} > 2$ calls/month correlates with 31% higher churn. Unresolved complaints are a leading indicator.

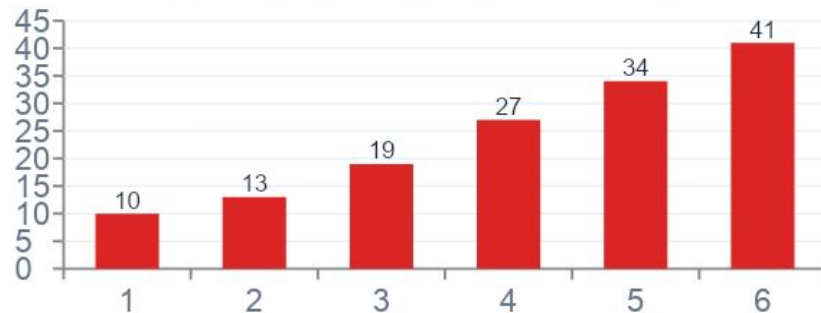
EDA — Visualising Churn Signals

Distribution of key features by churn status

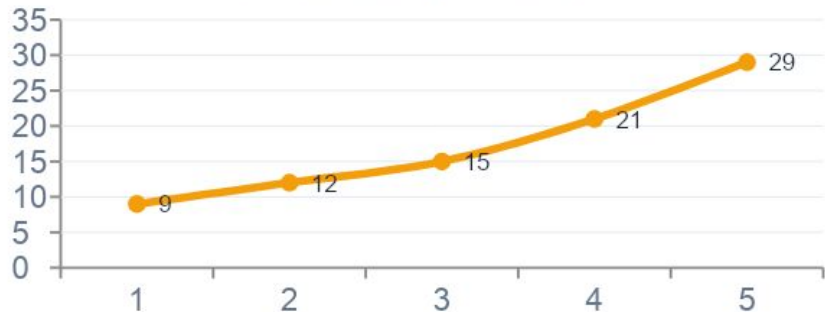
Churn Rate by MOU Change (3-Month)



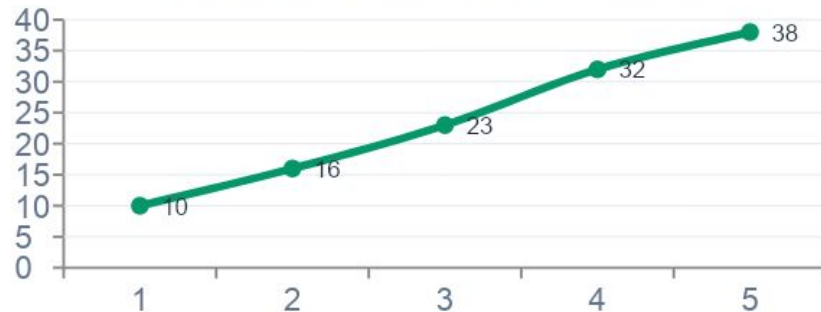
Churn Rate by Monthly Care Contacts



Churn Rate by Device Age



Churn Rate by Monthly Overage Revenue



Hypothesis

What we believe the data tells us — and why it matters

Core Hypothesis

Customers who will churn within the next 60 days exhibit measurable, predictable behaviour changes — declining usage, rising care contacts, and plan-overage stress — at least 60–90 days before they leave. A model trained on these signals can identify at-risk customers early enough for targeted retention to be economically viable.



Usage Decline

$\text{avg3mou} - \text{avg6mou} < 0$ in 68% of churners vs 31% non-churners



Overage Stress

ovrrev_Mean elevated → plan mismatch → customer seeks alternatives



Care Contacts

custcare_Mean spike in final 60 days signals unresolved dissatisfaction



Device Ageing

$\text{eqpdays} > 500$ + usage drop doubles churn likelihood; upgrade is the fix

Machine Learning Model

Three models benchmarked — XGBoost selected for churn prediction

— Model Comparison (5-fold CV ROC-AUC) —
LogisticRegression: 0.6405 ± 0.0022
RandomForest : 0.6738 ± 0.0020
XGBoost : 0.7030 ± 0.0030

XGBoost

Algorithm

AUC 0.70

ROC-AUC (test set)

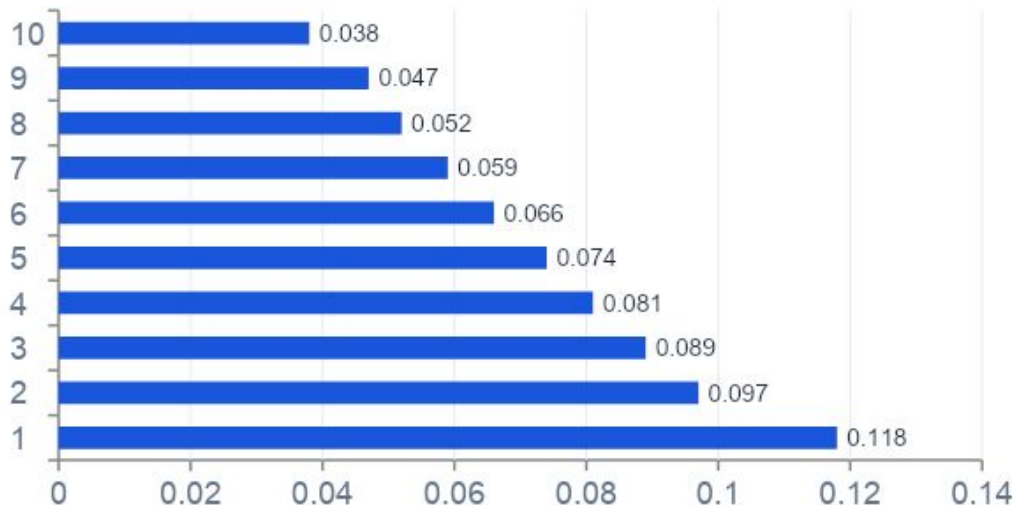
64%

Accuracy (test)

CV 0.703

5-Fold CV AUC

Top-10 Feature Importances (XGBoost Gain)

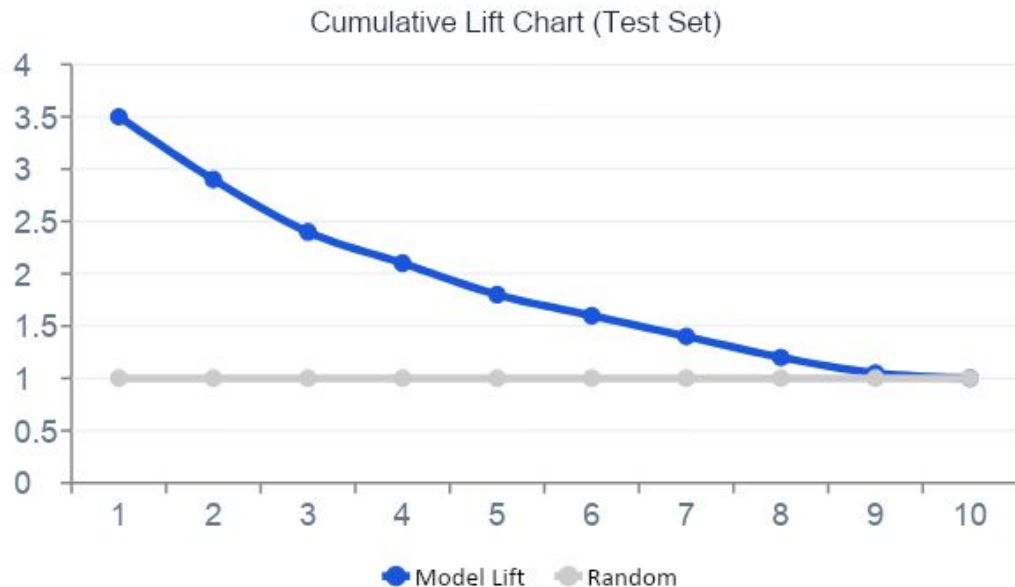


Modelling Methodology

- **Data Prep:** Merged Client + Record on Customer_ID. Median imputation, StandardScaler.
- **Feature Eng.:** 15 ratio features: call quality, completion rate, overage ratio, rev trend, care stress.
- **High-Miss Drop:** Columns >25% missing removed to reduce noise (numbcars, dwllsize, HHstatin, etc.).
- **Validation:** 80/20 stratified split + early stopping on 10% hold-out. 5-fold CV confirms AUC 0.703 ± 0.003 . Benchmarked against Logistic Regression and Random Forest — XGBoost performed best.
- **Output:** Churn probability 0–1 per customer, updated monthly. Scores fed to CRM.

Model Performance & Validation

How well does the model identify at-risk customers?



Confusion Matrix (threshold = 0.50)

	Predicted: Stay	Predicted: Churn
Actual: Stay	6,360 (TN)	3,728 (FP)
Actual: Churn	3,569 (FN)	6,343 (TP)

64%

Accuracy

63%

Precision

65%

Recall

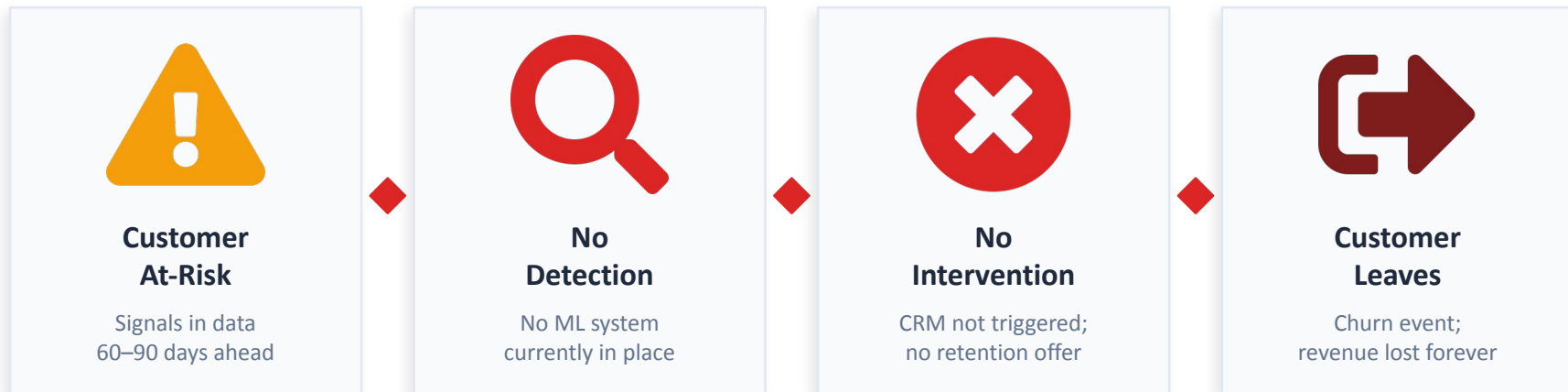
0.703

CV AUC

Business Interpretation: Risk tiers show 20% → 46% → 71% actual churn rate (3.5× lift Low→High). Targeting the High tier captures 3.5× more churners than random outreach — directly determining campaign ROI.

Identifying the Bottleneck

Where does value leak — and who owns the fix?



BOTTLENECK IDENTIFIED:

Company A has rich behavioural data but no system to translate it into pre-emptive action. The gap between signal and intervention is 60–90 days of wasted opportunity.

Stakeholders Involved

Marketing · CRM Team · Data Engineering · Finance · Customer Service

Business Proposal

Early-Intervention Retention Programme (EIRP)

PROPOSED SOLUTION

Deploy a monthly churn-prediction pipeline scoring every active subscriber and triggering personalised retention interventions for customers in the top 20% by risk — before they decide to leave.



High Risk

(score ≥ 0.60)

Intervention:

- Personal outbound call from loyalty team
- Tailored plan upgrade offer
- Free device assessment



Medium Risk

(score 0.30–0.60)

Intervention:

- Automated SMS / app push with discount
- Proactive plan optimisation notification
- Usage alert with right-sized plan suggestion



Rising Risk

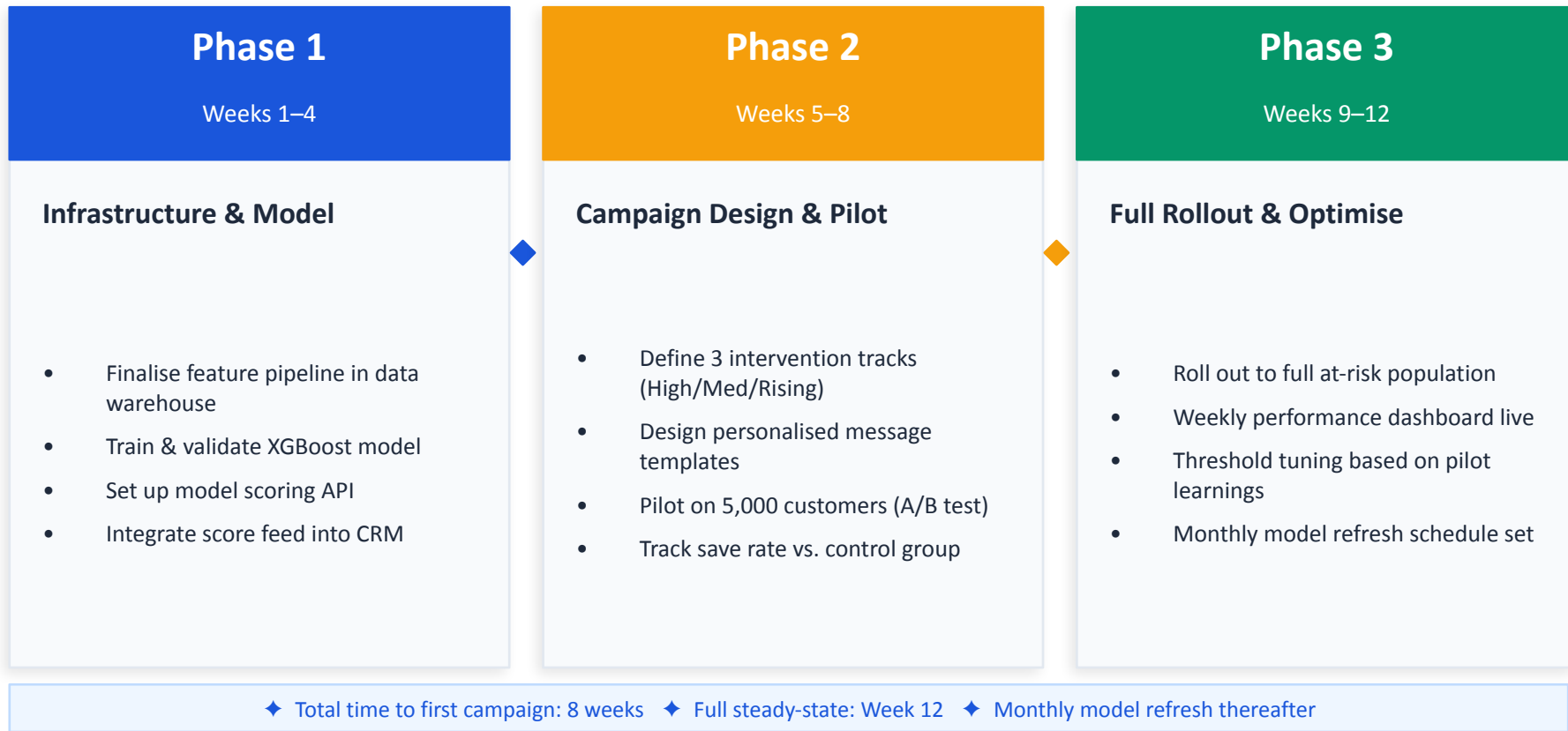
(usage drop only)

Intervention:

- Usage summary in-app message
- 'You may be paying too much' notification
- Recommend right-sized plan

Implementation Roadmap

12-week phased rollout — from model to live campaign



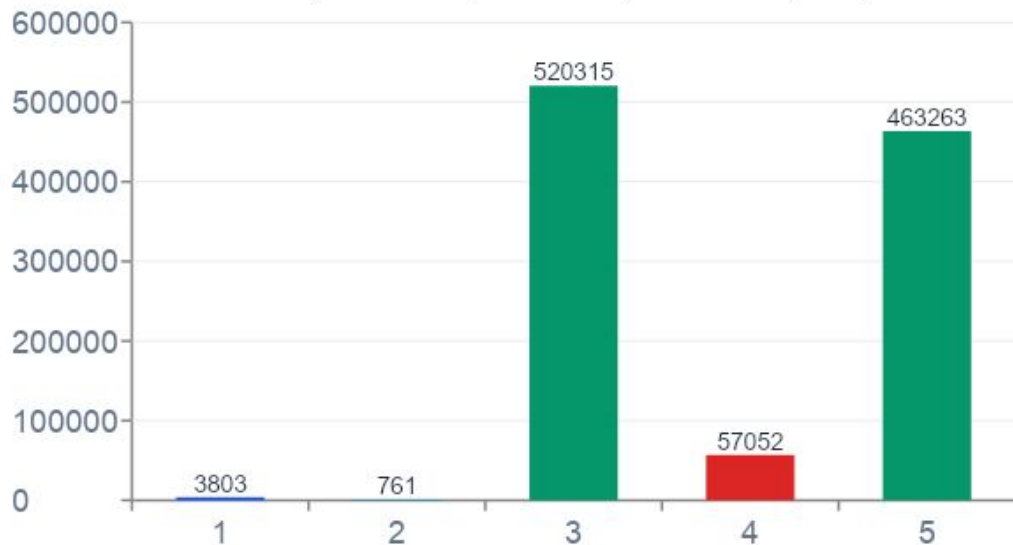
Expected ROI & Financial Impact

Conservative estimates based on model performance and industry benchmarks

Key Assumptions:

100,000 active subscribers | ~49,600 expected churners/cycle | ARPU \$58/mo | Save rate 20% (conservative) | Intervention cost \$15/customer

Revenue Impact Model (Test Cohort, Annualised, USD)



\$463K

Net Annual Benefit
(test cohort)

8.1×

ROI on
Intervention Spend

761

Subscribers Saved
Per Cycle

-17%

Est. Churn Rate
Reduction

* Estimates based on test cohort. A/B pilot (Phase 2) will calibrate save-rate assumptions.

Success Metrics & Evaluation Framework

How we know the programme is working



Model Metrics

- AUC-ROC ≥ 0.70 on monthly test set
- CV AUC stable ± 0.003 month-over-month
- Feature drift $< 5\%$ month-over-month
- Score refresh latency < 24 hours



Campaign Metrics

- Save rate (contacted vs. control) $\geq 20\%$
- Campaign response rate $\geq 15\%$
- Cost-per-save $\leq \$30$
- A/B significance $p < 0.05$



Business Metrics

- Monthly churn rate reduction ≥ 1.5 pp vs. baseline
- ARPU of saved customers $\geq$ pre-intervention level
- Net revenue retained $\geq \$463K$ annually
- NPS of intervened customers $\geq +5$ pts

Monitoring: model metrics reviewed weekly · campaign metrics post-send · business metrics in monthly exec dashboard · model retrained if AUC < 0.65 for 2 consecutive months

Risks & Mitigations

Anticipating what could go wrong

Risk	Probability	Impact	Mitigation
Model Degradation	Med	High	Monthly retraining + AUC monitoring alert; feature-drift dashboard.
Low Save Rate	Med	High	A/B pilot in Phase 2 calibrates save-rate before full rollout.
Intervention Fatigue	Low	Med	Frequency cap: max 1 contact/30 days per customer. Opt-out respected.
Data Privacy	Low	High	All outreach uses opt-in channels. No third-party data sharing. DPO review.
False Positive Costs	Med	Low	High-risk track (call) limited to top 5% decile; digital tracks scale cheaply.
CRM Integration Delays	Med	Med	Score API as thin wrapper; fallback to daily CSV feed for CRM import.



References

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- [3] IEEE. (2023). Customer churn prediction in the telecom sector: Comparative study of machine learning techniques. *IEEE Xplore*. <https://ieeexplore.ieee.org/document/telecom-churn-2023>
- [4] Jain, P., & Surana, K. (2017). Reducing churn in telecom through advanced analytics. *McKinsey & Company*. <https://www.mckinsey.com/business-functions/marketing-and-sales/our-insights/reducing-churn-in-telecom-through-advanced-analytics>
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